

# Study Design Document: Study II

## Task Switching comfort using colored light

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## Purpose of study (research hypothesis/questions)

The purpose of this study is to investigate how color-coded background light can influence user experience and comfort during cognitive task switching.

### Research questions are formulated as follows:

- I. How can cognitive task switching be supported by providing environments differentiated by color?
- II. Do differently colored environments lead to a more enjoyable work setting in multitasking scenarios?
- III. How does the comfortability of users change with introducing color-based work environments?

### The research hypotheses are structured as follows:

- I. Participants will report lower levels of perceived workload under coloured lighting compared to the uniform white lighting.
  - II. Participants will report higher comfort levels under coloured lighting compared to the uniform white lighting.
  - III. Coloured lighting will improve task performance compared to the uniform white lighting.
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## Participant profile

Participants are selected based on these criteria:

- **Inclusion criteria**

- Participants must have basic PC and keyboard experience, sufficient English language skills to read and type a text, and be older than 18 years old.
- Participants must have normal eyesight or have artifacts which allow for them to experience “normal” eyesight.

- **Exclusion criteria**

- Participants who self-report a history of photosensitive epilepsy or seizures triggered by visual stimuli will be excluded.
- Participants who self-report color blindness or significant visual impairments that are not corrected.

**Before entering the study potential participants will be provided with the following information via an information sheet/the online sign up site:**

This study involves working at a computer with colored LED lighting. The light may change color occasionally but will not strobe rapidly.

If you are a person with color blindness, have a history of photosensitive epilepsy, seizures triggered by visual stimuli, or any condition that may be aggravated by changing lights, you are advised not to participate in this study.

Participation is voluntary and can be discontinued any time before or during a study (participants will also be informed of this through an informed consent form), all collected data will be stored anonymously and used only for this study.

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## Compensation plan

- We participate in other students' studies.
  - Additionally, light refreshments (e.g. sweets, etc.) will be provided. No monetary compensation will be offered.
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## Methodology

### Study Design

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| <b>Qualitative</b> | <b>(Quantitative)</b><br>(specified in Study Design Document 1) |
|--------------------|-----------------------------------------------------------------|

### Experiments

|                       |                         |
|-----------------------|-------------------------|
| <b>Within-subject</b> | <b>Randomized order</b> |
|-----------------------|-------------------------|

|                                         |           |
|-----------------------------------------|-----------|
| <b>Estimated number of participants</b> | <b>20</b> |
|-----------------------------------------|-----------|

We use a within-subject experiment with one IV (light condition), where each participant completes 2 series of tasks: one under coloured light and one under baseline light. The order of conditions is counterbalanced and “randomly” assigned across participants to control learning effect, and general order effect. Participants will be able to take a break in between conditions to avoid fatigue. In each series one main task and 2 secondary tasks are performed to simulate cognitive task switching.

## Variables

| Independent variables                                                                                                                                  | Dependent variables                                                                                                    | Confounding variables                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Light condition:</b><br>(baseline: white light for all task switching; coloured light: main task with white light, secondary tasks with blue/green) | <b>Subjective comfort and flow:</b> assesses state flow<br><br>(Qualitative variables are defined in Study Document 1) | <b>Individual task switching ability:</b> some participants may feel task switching easier than others<br><br><b>Rest/break time:</b> fatigue can affect performance if one condition is always experienced at the same position<br><br><b>Pre-existing mental/neurological disorders:</b> Schizophrenia, ADHD, Alzheimer, colour blindness, visual impairments, etc. |

## Material & Apparatus

### Technical setup

- Laptop
- LED light with multiple colors and (remote) control
- Desk & chair
- Experiment Suite Software

### Task Setup

- Tasks to be performed by users

- **Main task:** Transcribe “The Adventure of Three Students” by Doyle
- **Smoke Task A:** Identify consonant/vowel vs. odd/even
- **Smoke Task B:** Spot-the-difference

## Color Setup

| Task         | Color | Light Bulb Setting |
|--------------|-------|--------------------|
| Main         | White | Normal             |
| Smoke Task A | Blue  | Brightest          |
| Smoke Task B | Green | Brightest          |

## Interview

A brief semi-structured interview (≈10-20 min) will be conducted after each participant's second block. The interview guide will follow a structured format: introduction, core questions (lighting & environment; task switching; flow; comfort), probing follow-ups, and a closing question. The guide was developed following Kallio et al. (2016) five-phase framework for interview guide development.

Interviews will be video- and audio-recorded on the Interviewees phones, transcribed and analysed.

[Non-visual colour: A qualitative study of how the totally blind and an achromatope navigate colour in the sighted world - ScienceDirect](#)

[Systematic methodological review: developing a framework for a qualitative semi-structured interview guide - PubMed](#)

### Semi-Structured Interview Guideline:

Intro Questions:

- How did you experience the tasks overall today?

Light:

- What did you notice about the lighting during the tasks in each condition?
- Do you remember the different light colors?
- How did you feel under coloured light?
- How did you feel under white light?

Comfort and well-being:

- Can you describe any moment when the environment felt especially comfortable or uncomfortable?

- How did you feel?
  - In what context?
  - How did the light setting contribute to that?
  - Would you prefer one lighting condition over the other for doing this kind of task? Please explain.

Flow:

- Optional - only if flow was not mentioned yet:
  - If you compare the two lighting conditions, in which did you feel more 'in the zone' and why?
- Were there moments when you felt 'lost' or had to pause to reorient yourself? What triggered that?

Words to look out for:

- Rushed
- Annoyed
- Distracted
- Surprised
- Supported
- Flow
- Comfortable
- Easier/Harder
- Overstimulated
- Faster/Slower → time related terms
- Prettier/... Aesthetic terms

Follow-up Questions:

- Can you tell me more about that moment?
- What exactly made you feel that way?
- How did that compare to the other condition?
- Was there anything unexpected about your feeling or performance?

Closing Question:

- Is there anything else you'd like to add about your experience with the tasks or the lighting conditions?

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## Structure & Timeline

**We will run the qualitative study together with the quantitative study.** The interview process will happen after the quantitative study.

The timeline therefore matches the timeline from the first study document, with the addition of the qualitative process.

Preparation required before the study starts:

- Informed consent form
- Demographic data
- Assessment of influential factors e.g. experience/expertise, drinking/smoking behavior

### Timeline of each task

| Task | Subject                                                                         | Time  |
|------|---------------------------------------------------------------------------------|-------|
| Main | Initial familiarity period.<br>Participants get comfortable with the main task. | 2min  |
| B    | Participants are asked to switch to smoke task B.                               | 30sec |
| Main | Participants are asked to switch back to the main task.                         | 30sec |
| A    | Participants are asked to switch to smoke task A.                               | 30sec |
| Main | Participants are asked to switch back to the main task.                         | 30sec |
| B    | Participants are asked to switch to smoke task B.                               | 30sec |
| Main | Participants are asked to switch back to the main task.                         | 30sec |
| A    | Participants are asked to switch to smoke task A.                               | 30sec |
| Main | Participants are asked to switch back to the main task.                         | 30sec |
| Q    | Questionnaire + data saving                                                     | 2min  |
|      | Total                                                                           | 8min  |

### Timeline of experiment

- Quantitative part
  - Participants get introduced to the study
  - Participants familiarize themselves with the task
  - Break
  - Participants complete the first iteration of the tasks and complete a questionnaire

- Break
- Participants complete the second iteration of the tasks and complete a questionnaire
- Qualitative part
  - Break
  - Participants get interviewed by us

| Phase       | Subject                                                                                                                                                                                                                                                                                                                                                                  | Material                                     | Time                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------|
| Prep        | Participants will be informed about the experiment and its structure and will sign a consent form (informed consent form produced by the Frankfurt University of Applied Sciences - <a href="https://hci-studies.org/informed-consent-generator/">https://hci-studies.org/informed-consent-generator/</a> )<br>Collection of demographic data and "influential factors". | Consent form                                 | 2min                |
|             | Explanation round <ul style="list-style-type: none"> <li>● Explain main task</li> <li>● Explain smoke tasks</li> </ul>                                                                                                                                                                                                                                                   |                                              | 3min                |
|             | Let participants figure out the mechanics and ways of interaction of the tasks.                                                                                                                                                                                                                                                                                          |                                              | 3min<br>(1min each) |
|             | Break                                                                                                                                                                                                                                                                                                                                                                    |                                              | 3min                |
| Iteration 1 | Task iteration 1.<br>Randomly assigned with-light or without-light group.                                                                                                                                                                                                                                                                                                | Light setup<br>Laptop setup<br>Questionnaire | 8min                |
|             | Break                                                                                                                                                                                                                                                                                                                                                                    |                                              | 3min                |
| Iteration 2 | Task iteration 2.<br>Inverse group assignment.                                                                                                                                                                                                                                                                                                                           | Light setup<br>Laptop setup<br>Questionnaire | 8min                |
|             | Break                                                                                                                                                                                                                                                                                                                                                                    |                                              | 3min                |
|             | Interview                                                                                                                                                                                                                                                                                                                                                                | Interview guideline                          | 12min               |
|             |                                                                                                                                                                                                                                                                                                                                                                          | Total                                        | 45min               |



# Analysis

## Qualitative Analysis

Interview data will be analysed using thematic analysis to identify recurring patterns related to lighting experience, task switching, flow, and comfort. Themes will be generated inductively from the transcripts and used to complement and contextualize the quantitative findings.

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## Personnel Involved & Responsibilities

For better time efficiency, everyone has two roles and every role is included twice so that we can run the study in parallel. Every team member is vetted on how the study is conducted and how the Software works and is thus able to conduct it in full.

|          |                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Celina   | Conduct Study <ul style="list-style-type: none"><li>- Receptionist (consent form provider, prepare participants, etc.)</li><li>- Light operator</li></ul> |
| Kseniia  | Conduct Study <ul style="list-style-type: none"><li>- Receptionist (consent form provider, prepare participants, etc.)</li><li>- Light operator</li></ul> |
| Sepehr   | Conduct Study <ul style="list-style-type: none"><li>- Light operator</li></ul>                                                                            |
| Omar     | Conduct Study <ul style="list-style-type: none"><li>- Receptionist (consent form provider, prepare participants, etc.)</li><li>- Light operator</li></ul> |
| Johannes | Conduct Study <ul style="list-style-type: none"><li>- Software implementation and handling</li><li>- Light operator</li></ul>                             |